Detecting Seizure Onset Patterns as Temporal Objects

A Generalizable YOLO-Based Framework for iEEG Time Series Event Detection

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Goals

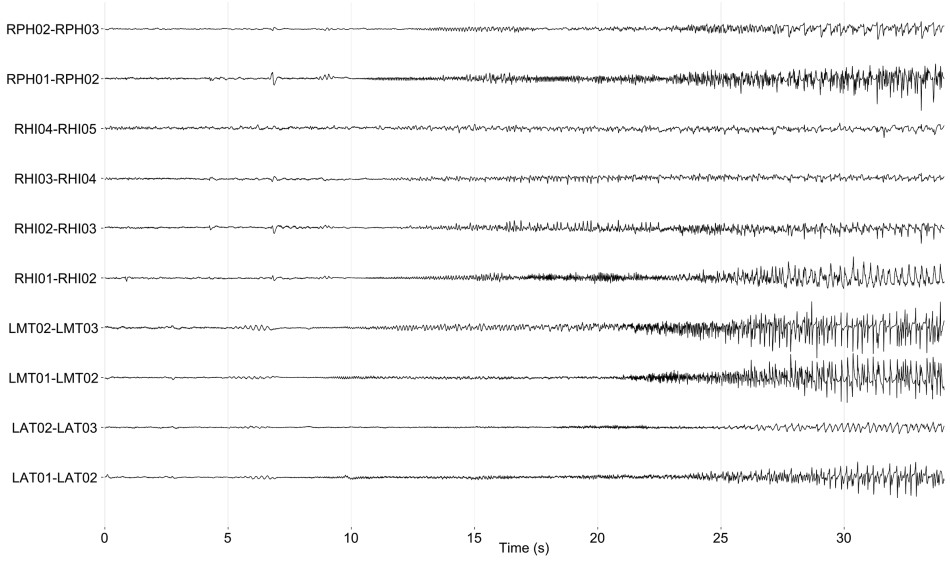
- Develop a generalizable method to detect multiple patterns in the EEG

- Automated, high-throughput, multi-class simultaneously
- Examples:
  - Seizure onset patterns
  - Slowing, rhythmic activity
  - Interictal bursts

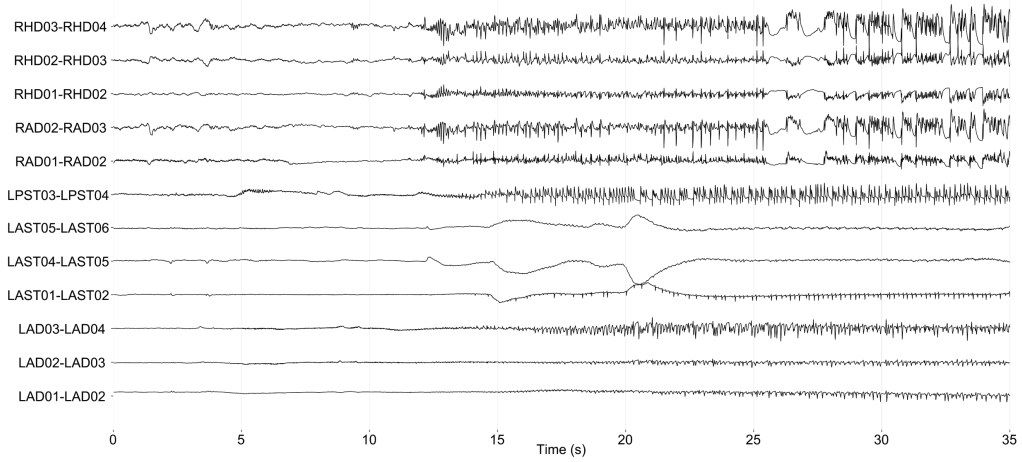
Ex: Seizure Onset Patterns - localization

- Inform seizure onset zone
- Predict surgical outcome

- BUT: automated detection is difficult
  - Different time scales
    <1 second vs > 30 sec
  - Many classes



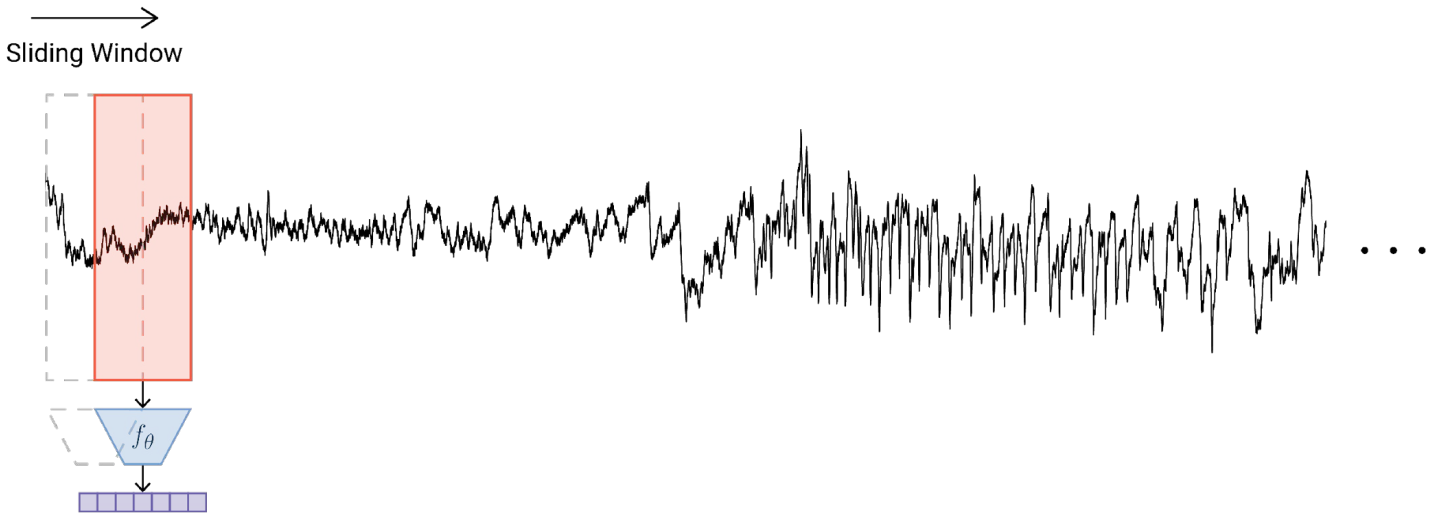
LVFA
Good Outcome



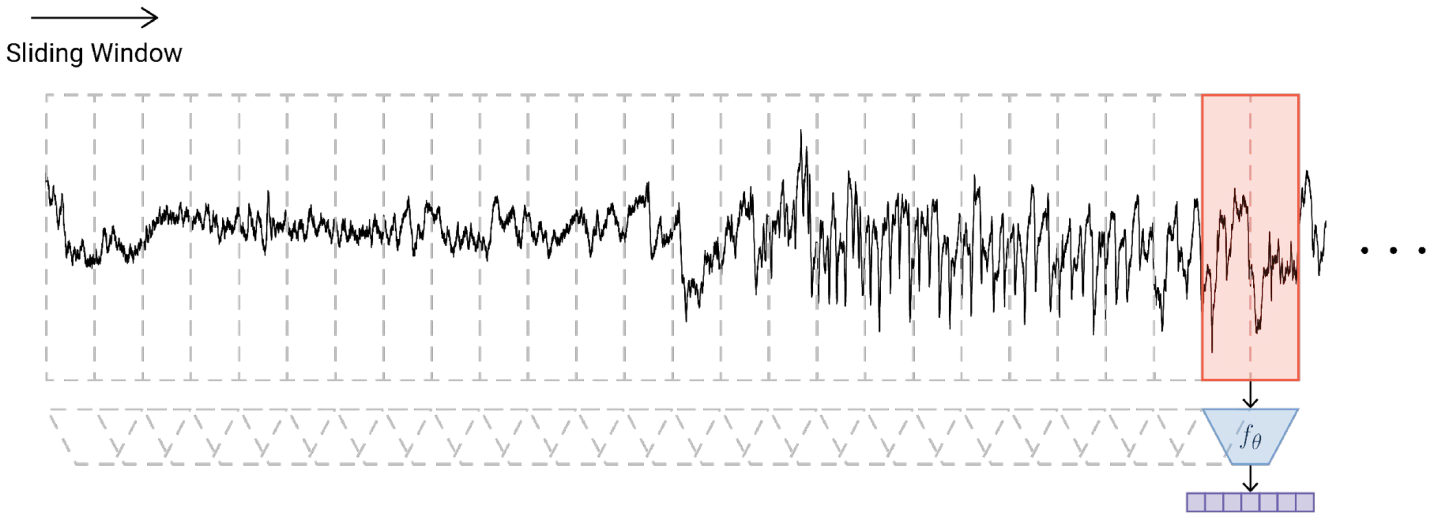
Repetitive Spiking
Poor Outcome

EDIT FIGURES

The current approach



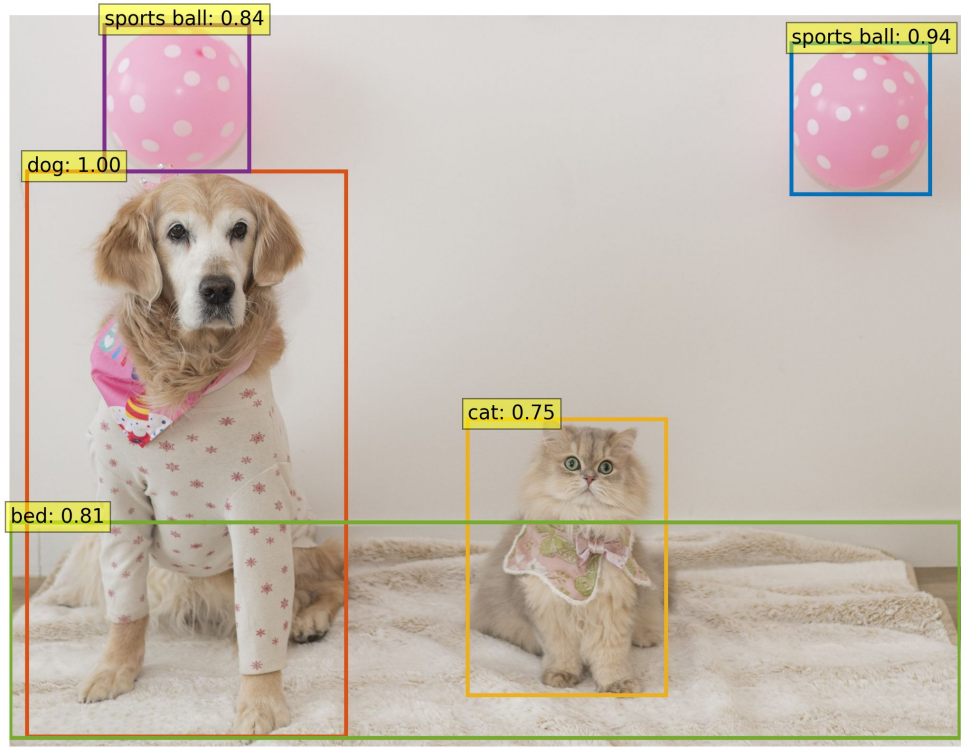
The current approach



TO EDIT

Our Approach: YOLO

- From computer vision
- Different sizes, classes that overlap
- 1D implemention



Experimental Flow

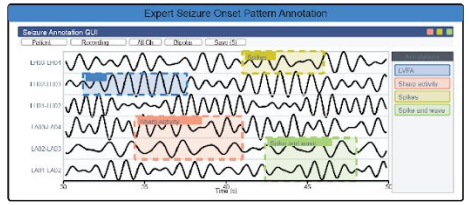
Patient Data

57 Patients
30F / 27M
Age: 20–70
Surgery: 49, Other: 8
Focal: 48, General: 9
Favorable (Engel@12m): 44

Expert SOP Annotation

Expert annotators draw *per-channel* bounding boxes via custom GUI using standardized labels

LVFA, Rhythmic, Sharp, Spike



Data Processing

Dataset

57 Patients
143 Seizures
1549 Annotations

Preprocessing

Bandpass Filter
Bipolar Montage
60s Sliding Windows

Model Training

Backbone
Feature extraction

Neck
FPN / PAN fusion

Head
Predict bbox + cls

LVFA
Spikes
S&W

Evaluation

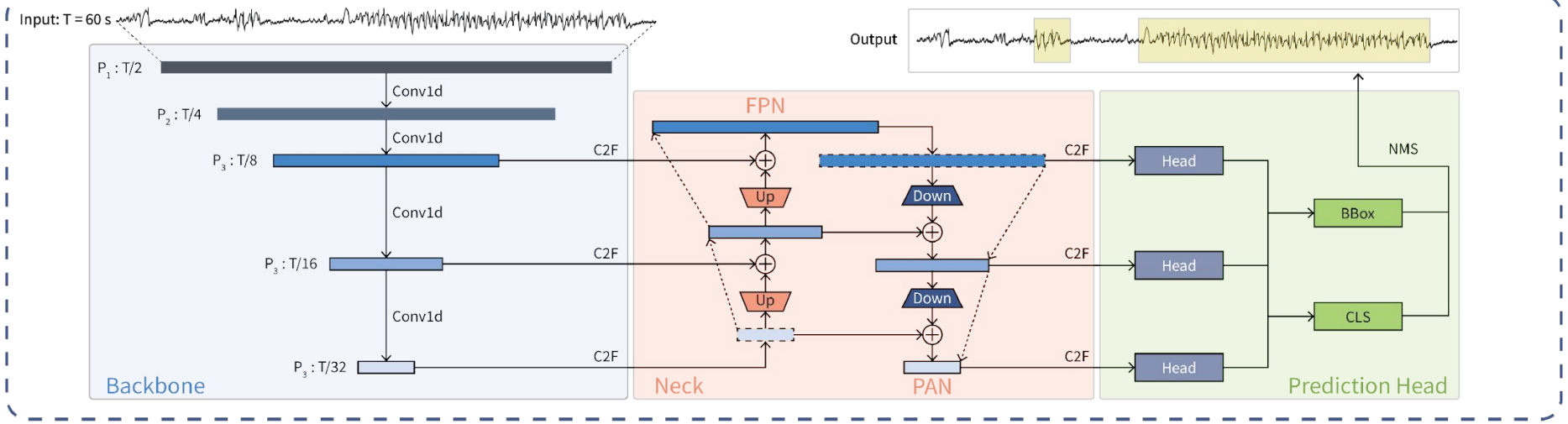
Leave-Patient-Out Cross Validation

Patient 1
Patient 2
Patient 3
Patient N [unseen]

External Dataset Interrater Analysis

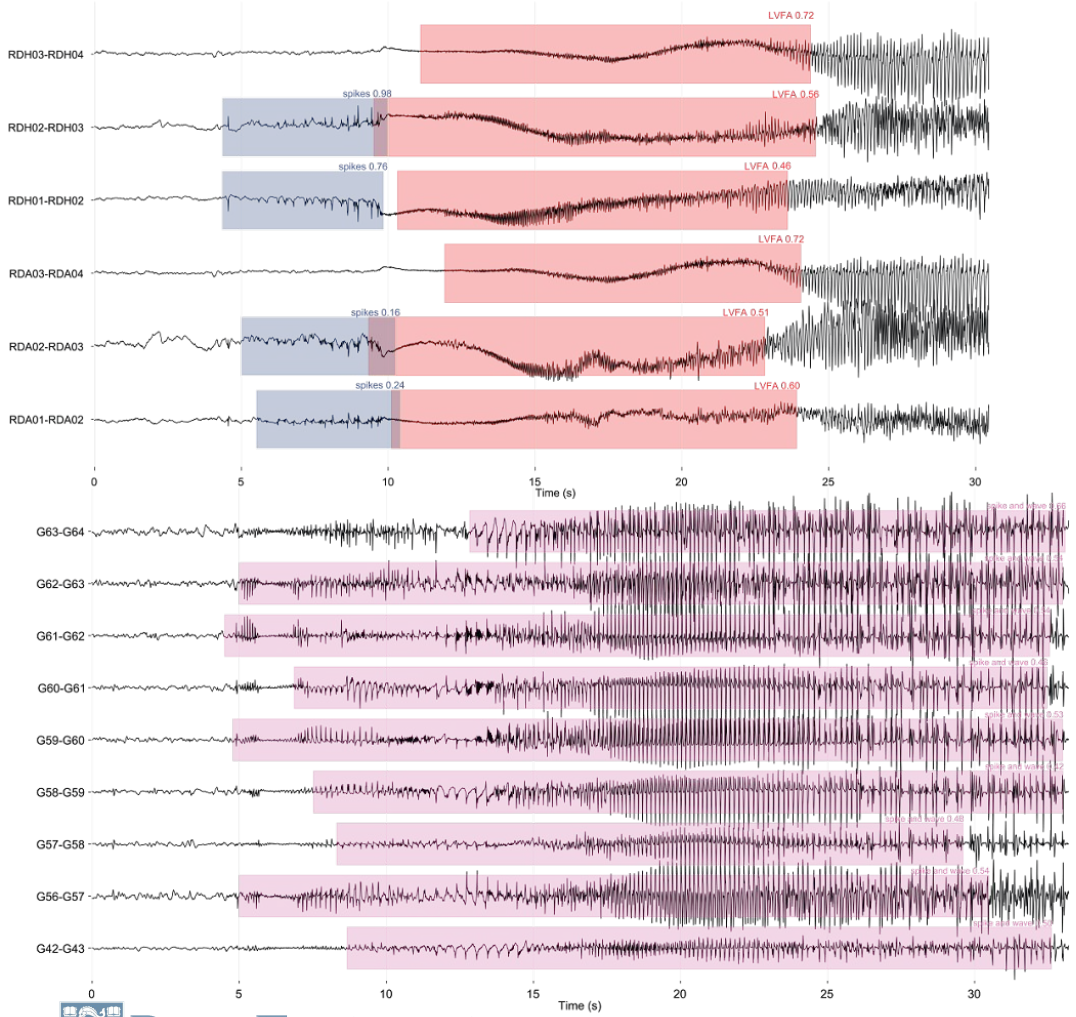
Onset time
SOP class

YOLOv8-1D



Results

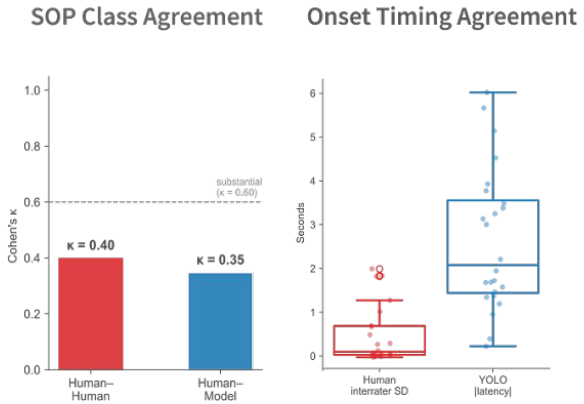
SOP Detection: Example Prediction



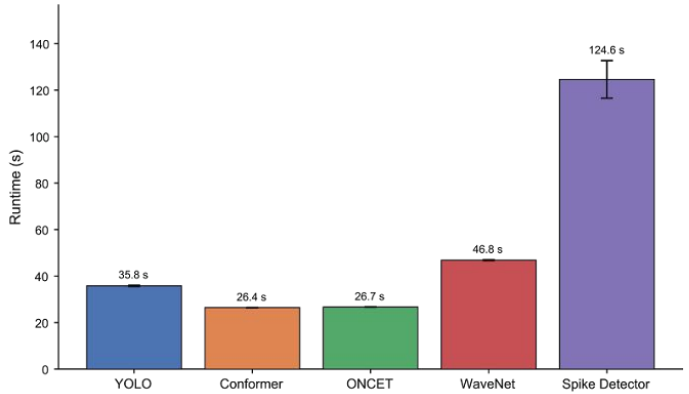
SOP Leave-Patient-Out Cross Validation Performance

Model	% Precision	% Recall	mAP@50	mAP@50-95
YOLOv8-1D (OURS)	0.635 ± 0.027	0.848 ± 0.022	0.683 ± 0.017	0.603 ± 0.021

Inter-Rater Reliability



Inference Speed



Generalizability: Spike & Seizure Detection

Model	% Precision	% Recall	mAP@50	mAP@50-95
Ours - spike & seizure	0.579 ± 0.002	0.648 ± 0.002	0.652 ± 0.001	0.526 ± 0.001
Spike Detector	0.712 ± 0.013	0.575 ± 0.013	n/a	n/a
Ours - spike only	0.576 ± 0.003	0.650 ± 0.002	0.614 ± 0.002	0.449 ± 0.002
WaveNet				
ONCET				
Ours - seizure only	0.596 ± 0.002	0.640 ± 0.002	0.691 ± 0.001	0.603 ± 0.002

Conclusions

- Novel tool for EEG pattern detection
- Comparable to experts
- MUCH faster
- Multiple patterns simultaneously
- Promising

<u>In Progress</u>:

- Improve Performance: More data, patterns, applications
- Publish code, data and model

Questions?